SAT UZBEKISTAN

TURBO-MODULE-2.23

30 QUESTION TURBO TEST

Focused Digital SAT Math Practice

@SATUzbekistan

Test creator Xodiyev Lutfulloxon 10 x (800/800)

THIS BOOKLET INCLUDES

30	Digital SAT Math questions
19	multiple-choice questions
11	student-produced response questions

1 Turbo-Module-2.23

Mark for Review

When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 8 resistors connected in series, where the resistance of each resistor is positive. This circuit has a total resistance of 130 ohms. The total resistance of 3 resistors in this circuit is 90 ohms. Which inequality best represents all possible values of the resistance x, in ohms, of one of the other 5 resistors in this circuit?

A. $0 < x < 8$

B. $0 < x < 40$

C. $40 < x < 90$

D. $40 < x < 130$

1-masala yangi terminlar

resistor - rezistor
circuit - elektr zanjiri
connected in series - ketma-ket ulangan
resistance - qarshilik
total resistance - umumiy qarshilik
sum of the resistances - qarshiliklar yig‘indisi
positive resistance - musbat qarshilik
inequality - tengsizlik
possible values - mumkin bo‘lgan qiymatlar
ohms - om

2

Turbo-Module-2.23

Mark for Review

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of 47/45 ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b?

A) 47

B) 4

C) $\frac{47}{45}$

D) $\frac{4}{45}$

2-masala yangi terminlar

engineer - muhandis
resistors in series - ketma-ket ulangan rezistorlar
total resistance - umumiy qarshilik
type A resistor - A turdagi rezistor
type B resistor - B turdagi rezistor
resistance - qarshilik
given equation - berilgan tenglama
positive difference - musbat farq
value of a - a ning qiymati
value of b - b ning qiymati

3

Turbo-Module-2.23

Mark for Review

Several resistors are connected in series. A circuit has 9 resistors with positive resistance. The total resistance of 4 resistors is 60 ohms, and each of the other 5 resistors has resistance at most 25 ohms. If the total resistance is x ohms, which inequality best represents the situation?

A) $0 < x \leq 185$

B) $60 < x \leq 125$

C) $60 < x \leq 185$

D) $x \geq 125$

3-masala yangi terminlar

resistors in series - ketma-ket ulangan rezistorlar
circuit - elektr zanjiri
positive resistance - musbat qarshilik
total resistance - umumiy qarshilik
other resistors - qolgan rezistorlar
at most - ko‘pi bilan
inequality - tengsizlik
possible values - mumkin bo‘lgan qiymatlar
ohms - om

4 Turbo-Module-2.23

Mark for Review

For a particular car, the linear function f gives the predicted power, in brake horsepower (bhp), for engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm. According to this function, the car's predicted power is 228 bhp at an engine speed of 1,896 rpm and 600 bhp at an engine speed of 4,500 rpm. The equation defines f, where x is the engine speed, in rpm, and a is a constant. What is the value of a?

$$f(x) = \frac{1}{7}(x - a) + 228$$

Enter your answer

4-masala yangi terminlar

linear function - chiziqli funksiya
predicted power - bashorat qilingan quvvat
brake horsepower - tormoz ot kuchi
engine speed - dvigatel tezligi
revolutions per minute - daqiqadagi aylanishlar
rpm - aylanish/minut
equation defines f - f ni aniqlovchi tenglama
constant - oʻzgarmas
value of a - a ning qiymati

5 Turbo-Module-2.23

Mark for Review

A right square pyramid has a total surface area of 28,160 square inches, and the combined surface area of the four lateral faces of this pyramid is 16,060 square inches. What is the height, in inches, of this pyramid?

A) 48

B) 55

C) 73

D) 110

5-masala yangi terminlar

right square pyramid - asosi kvadrat bo‘lgan to‘g‘ri piramida
total surface area - to‘la sirt yuzi
lateral faces - yon yoqlar
combined surface area - umumiy yon sirt yuzi
base - asos
height - balandlik
square inches - kvadrat dyuym
pyramid - piramida

6 Turbo-Module-2.23

Mark for Review

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is 100 cm^2. Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2?

A) 209

B) 220

C) 367

D) 551

6-masala yangi terminlar

right square pyramid - asosi kvadrat bo‘lgan to‘g‘ri piramida
first pyramid - birinchi piramida
second pyramid - ikkinchi piramida
height - balandlik
base area - asos yuzasi
surface area - sirt yuzi
difference - farq
square centimeters - kvadrat santimetr
closest to - eng yaqin

7

Turbo-Module-2.23

Mark for Review

Two cylinders, A and B, have congruent bases. Cylinder A has a radius of 4 inches, and the height of cylinder B is half the height of cylinder A. The surface area of cylinder A is 144π square inches. Cylinders A and B are joined by gluing one base of cylinder A to one base of cylinder B, forming a new cylinder. What is the height, in inches, of the new cylinder?

Enter your answer

7-masala yangi terminlar

cylinder - silindr
congruent bases - teng asoslar
radius - radius
height - balandlik
half the height - balandlikning yarmi
surface area - sirt yuzi
square inches - kvadrat dyuym
joined - birlashtirilgan
gluing one base - bir asosni yopishtirish
new cylinder - yangi silindr

8 Turbo-Module-2.23

Mark for Review

Arturo builds and sells two types of steel tables. The revenue Arturo generates from selling each square table is 2 times his cost to build a square table, and the revenue Arturo generates from selling each hexagonal table is 2.5 times his cost to build a hexagonal table. In one year, Arturo's total cost to build tables was $25,740, and he generated a total revenue of $61,875 from selling all the tables he built that year. For that year, how much greater is the revenue Arturo generated from selling hexagonal tables than the revenue he generated from selling square tables?

Enter your answer

______

8-masala yangi terminlar

revenue - daromad
steel tables - po'lat stollar
square table - kvadrat stol
hexagonal table - olti burchakli stol
cost to build - ishlab chiqarish xarajati
total cost - jami xarajat
total revenue - jami daromad
generated from selling - sotishdan olingan
times the cost - xarajatga nisbatan baravar
how much greater - qancha ko'p

9 Turbo-Module-2.23

Mark for Review

$$50x + 49y = c, \qquad 49x - 50y = c$$

In the given system of equations, c is a positive constant. Which of the following could be the point where the graphs of the equations in this system intersect in the xy-plane?

A) $(10, -\frac{10}{99})$

B) $(10, 990)$

C) $(c, \frac{c}{99})$

D) $(c, 0)$

9-masala yangi terminlar

system of equations - tenglamalar sistemasi
positive constant - musbat oʻzgarmas
graphs of equations - tenglamalar grafiklari
intersect - kesishmoq
point of intersection - kesishish nuqtasi
xy-plane - koordinata tekisligi
ordered pair - tartiblangan juftlik
possible point - mumkin boʻlgan nuqta

10 Turbo-Module-2.23

Mark for Review

In triangle JKL, the measure of angle J is $(90b)°$, the measure of angle K is $(71a)°$, and the measure of angle L is $(19a)°$, where a and b are constants. Which of the following must be true?

A) $\cos L > \sin K$

B) $\cos L = \sin K$

C) $\cos L < \sin K$

D) There is not enough information to compare the values of cos L and sin K.

10-masala yangi terminlar

triangle - uchburchak
measure of angle - burchak o'lchovi
degrees - graduslar
constants - o'zgarmaslar
cosine - kosinus
sine - sinus
compare - taqqoslamoq
must be true - albatta to'g'ri
not enough information - ma'lumot yetarli emas

11

Turbo-Module-2.23

Mark for Review

The expression $6x^4 + 17x^2 + 7$ can be written as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Enter your answer

11-masala yangi terminlar

expression - ifoda
can be written as - ... ko‘rinishda yozilishi mumkin
positive integers - musbat butun sonlar
positive nonintegers - musbat butun bo‘lmagan sonlar
factorization - ko‘paytuvchilarga ajratish
factor - ko‘paytuvchi
coefficient - koeffitsiyent
value of a + c - a + c ning qiymati

12 Turbo-Module-2.23

Mark for Review

$$8x^2 + 112px + 8y^2 - 64py = -448p^2$$

In the xy-plane, the graph of the given equation is a circle. The length of the radius of the circle is np, where n and p are positive constants. What is the value of n?

Enter your answer

12-masala yangi terminlar

xy-plane - koordinata tekisligi
graph of an equation - tenglama grafigi
circle - aylana
equation of a circle - aylana tenglamasi
radius - radius
length of the radius - radius uzunligi
positive constants - musbat oʻzgarmaslar
value of n - n ning qiymati

13

Turbo-Module-2.23

Mark for Review

A raised garden is in the shape of a right rectangular prism. Its base has a width of 3 feet and a length of 27 feet, and it will be filled 24 inches high with topsoil. The total cost of the topsoil needed to fill the garden to this height is 234 dollars. What is the unit cost, in dollars per cubic yard, for the topsoil? (1 yard = 3 feet; 1 foot = 12 inches)

Enter your answer

13-masala yangi terminlar

raised garden - baland ko‘tarilgan ekin maydoni
right rectangular prism - to‘g‘ri to‘rtburchakli prizma
base - asos
width - kenglik
length - uzunlik
filled high - balandlikkacha to‘ldirilgan
topsoil - unumdor tuproq
total cost - jami xarajat
unit cost - birlik narxi
cubic yard - kub yard
unit conversion - birlik almashtirish

14 Turbo-Module-2.23

Mark for Review

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 2 feet above the ground and reaches its maximum height of 258 feet above the ground 4 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 5 seconds after being launched?

A) 58

B) 194

C) 242

D) 254

14-masala yangi terminlar

quadratic function - kvadrat funksiya
models the height - balandlikni modellashtiradi
height - balandlik
feet - fut
time in seconds - sekundlardagi vaqt
launched - uchirilgan
elevated surface - baland sirt
initial height - boshlang‘ich balandlik
maximum height - eng katta balandlik
model - model

15 Turbo-Module-2.23

Mark for Review

$$k(x) = rx^2 + 8x + t$$

In the given quadratic function, r and t are constants. The graph of y = k(x) in the xy-plane is a parabola that opens downward and has a vertex at the point (u, v), where u and v are constants. If v > 0 and k(-1) = k(3) > 0, which of the following must be true?

I. $t > 0$
II. $r < -8$

A) I only

B) II only

C) I and II

D) Neither I nor II

15-masala yangi terminlar

quadratic function - kvadrat funksiya
constants - o‘zgarmaslar
graph - grafik
xy-plane - koordinata tekisligi
parabola - parabola
opens downward - pastga ochiladi
vertex - uch nuqta
positive value - musbat qiymat
symmetric inputs - simmetrik argumentlar
must be true - albatta to‘g‘ri

16

Turbo-Module-2.23

Mark for Review

Two similar right circular cylinders, cylinder A and cylinder B, are placed one on top of the other with their central axes aligned. Cylinder A is the smaller cylinder. Cylinder A has a radius of 6 inches and a height of 8 inches. Cylinder B has a radius of 9 inches. The circular base of cylinder A is centered on the top base of cylinder B. What is the total exterior surface area, in square inches, of the combined solid?

A) 378π

B) 414π

C) 450π

D) 474π

16-masala yangi terminlar

similar right circular cylinders - o‘xshash to‘g‘ri doiraviy silindrlar
cylinder A - A silindr
cylinder B - B silindr
central axes aligned - markaziy o‘qlar ustma-ust
radius - radius
height - balandlik
circular base - doiraviy asos
total exterior surface area - umumiy tashqi sirt yuzi
combined solid - birlashtirilgan jism
square inches - kvadrat dyuym

17 Turbo-Module-2.23

Mark for Review

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 62 sedans for sale at the dealership on Monday, how many minivans were for sale?

A) 93

B) 310

C) 9

D) 36

17-masala yangi terminlar

car dealership - avtosalon
sedans - sedanlar
SUVs - SUV avtomobillar
minivans - minivenlar
vehicles for sale - sotuvdagi avtomobillar
percent - foiz
total vehicles - jami avtomobillar
for sale - sotuvda
how many minivans - nechta miniven

18 Turbo-Module-2.23

Mark for Review

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

Angel Cruz	483
Terry Smith	320

A 163

B 1,304

C 3,864

D 5,621

18-masala yangi terminlar

poll - so'rovnoma
voters - saylovchilar
selected at random - tasodifiy tanlangan
candidate - nomzod
upcoming election - bo'lajak saylov
sample - tanlanma
expected votes - kutilayotgan ovozlar
expected to win - yutishi kutilmoqda
total voters - jami saylovchilar
by how many votes - nechta ovoz farqi bilan

19 Turbo-Module-2.23

Mark for Review

Kai used fabric measuring 4 yards in length to make each costume for a school play. The relationship between the number of costumes that Kai made, x, and the total length of fabric that he purchased, y, in yards, is represented by the equation $y - 4x = 5$. What is the best interpretation of 5 in this context?

A. Kai made 5 costumes.

B. Kai purchased a total of 5 yards of fabric.

C. Kai used a total of 5 yards of fabric to make the costumes.

D. Kai purchased 5 yards more fabric than he used to make costumes.

19-masala yangi terminlar

fabric - mato
yards in length - yardlarda uzunlik
costume - kostyum
number of costumes - kostyumlar soni
total length - umumiy uzunlik
purchased - sotib olgan
equation - tenglama
interpretation - talqin
in this context - shu kontekstda
yards more - yardga ko‘proq

20 Turbo-Module-2.23

Mark for Review

A small business owner budgets \$2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays \$4.90 per candle to purchase small candles and \$11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

Enter your answer

20-masala yangi terminlar

small business owner - kichik biznes egasi
budget - byudjet
purchase candles - sham sotib olmoq
minimum - eng kam
discounted pricing - chegirmali narx
price per candle - har bir sham narxi
small candles - kichik shamlar
large candles - katta shamlar
maximum number - eng ko‘p son
stay within the budget - byudjetdan oshmaslik

21 Turbo-Module-2.23

Mark for Review

An object's speed, in meters per hour, is increasing at a rate of 1,530 meters per hour per second. Which of the following is closest to this rate in feet per second squared? (Use 1 meter = 3.28 feet.)

A) 0.13

B) 1.39

C) 7.77

D) 83.6

21-masala yangi terminlar

speed - tezlik
meters per hour - soatiga metr
increasing at a rate - tezlik bilan ortmoqda
meters per hour per second - sekundiga soatiga metr
feet per second squared - sekund kvadratiga fut
unit conversion - birlik almashtirish
conversion factor - almashtirish koeffitsiyenti
closest to - eng yaqin
1 meter = 3.28 feet - 1 metr = 3.28 fut

22 Turbo-Module-2.23

Mark for Review

$$f(x) = (x - a)(x - b)(x + 53)$$

The function f(x) is defined by (x - a)(x - b)(x + 53), where a and b are constants. f(-50) > 0 and f(-48) < 0. When the function f(x) is divided by x + 45, the remainder is 0. If the function f(x) passes through the point (c, 0), where c is an integer constant, what is the largest possible value of c?

Enter your answer

22-masala yangi terminlar

function - funksiya
defined by - orqali aniqlangan
constants - o‘zgarmaslar
remainder - qoldiq
divided by - ga bo‘linganda
factor - ko‘paytuvchi
passes through a point - nuqtadan o‘tadi
x-intercept - x o‘qi bilan kesishish
integer constant - butun o‘zgarmas
largest possible value - eng katta mumkin bo‘lgan qiymat

23 Turbo-Module-2.23

Mark for Review

In the xy-plane, lines k and j are perpendicular and intersect at the point (5, 26). Line j passes through the origin, and line k passes through the point (p, 0), where p is a constant. What is the value of p?

A) -130.2

B) -129.8

C) 139.8

D) 140.2

23-masala yangi terminlar

xy-plane - koordinata tekisligi
lines - to‘g‘ri chiziqlar
perpendicular - perpendikulyar
intersect - kesishmoq
point of intersection - kesishish nuqtasi
origin - koordinata boshi
passes through - nuqtadan o‘tadi
constant - o‘zgarmas
slope - qiyalik
x-intercept - x o‘qi bilan kesishish

24

Turbo-Module-2.23

Mark for Review

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field's strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is 3 times the side length, in meters, of the smaller square. An electric field with strength 29.00 volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is 4,640 volts · meters, what is the electric flux, in volts · meters, of the electric field through the larger square?

Enter your answer

24-masala yangi terminlar

electric field - elektr maydoni
flat surface - tekis sirt
perpendicular - perpendikulyar
electric flux - elektr oqimi
field strength - maydon kuchlanganligi
area of the surface - sirt yuzasi
adjacent squares - yonma-yon kvadratlar
side length - tomon uzunligi
uniformly - bir tekis
total electric flux - umumiy elektr oqimi
volts per meter - volt/metr
volts · meters - volt · metr
larger square - katta kvadrat

25 Turbo-Module-2.23

Mark for Review

$$(x+6a)^2+(y-38a)^2=36a^2$$

In the xy-plane, circle M is the graph of the equation shown, where a is a positive constant. Circle V is obtained by shifting circle M 12a units to the right. Which equation represents circle V?

A. $(x+18a)^2+(y-38a)^2=36a^2$

B. $(x+6a)^2+(y-50a)^2=36a^2$

C. $(x+6a)^2+(y-26a)^2=36a^2$

D. $(x-6a)^2+(y-38a)^2=36a^2$

25-masala yangi terminlar

xy-plane - koordinata tekisligi
circle - aylana
graph of the equation - tenglama grafigi
positive constant - musbat o‘zgarmas
shifting - siljitish
units to the right - o‘ngga birlik
equation represents - tenglama ifodalaydi
center - markaz
radius - radius
translation - parallel ko‘chirish

26 Turbo-Module-2.23

Mark for Review

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is $14,000, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

A. $148.15

B. $161.53

C. $172.84

D. $184.94

26-masala yangi terminlar

business owner - biznes egasi
chair - stul
employees - xodimlar
total budget - umumiy byudjet
sales tax - savdo solig‘i
includes a 7% sales tax - 7% savdo solig‘ini o‘z ichiga oladi
maximum possible price - eng katta mumkin bo‘lgan narx
price per chair - har bir stul narxi
before sales tax - soliqdan oldin
closest to - eng yaqin

27 Turbo-Module-2.23

Mark for Review

For the function f, for each increase in the value of x by c, where c is a positive constant, the value of f(x) increases by a factor of 27. Which of the following equivalent forms of the function f displays 1/c as a coefficient of x?

A $f(x) = 48(3)^{\frac{1}{2}x}$

B $f(x) = 48(3^3)^{\frac{1}{6}x}$

C $f(x) = 48(9)^{\frac{1}{4}x}$

D $f(x) = 48(27^{\frac{1}{3}x})^{\frac{1}{2}}$

27-masala yangi terminlar

function - funksiya
increase in x - x ning oshishi
positive constant - musbat o‘zgarmas
increases by a factor of 27 - 27 marta ortadi
equivalent forms - teng kuchli shakllar
coefficient of x - x ning koeffitsiyenti
exponential function - eksponensial funksiya
exponential form - eksponensial shakl
factor of 27 - 27 ko‘payish koeffitsiyenti
displays 1/c - 1/c ni ko‘rsatadi

28 Turbo-Module-2.23

Mark for Review

$$(x+4)^2 + (y-19)^2 = 121$$

The graph of the given equation is a circle in the xy-plane. The point (a, b) lies on the circle. Which of the following is a possible value for a?

A -16

B -14

C 11

D 19

28-masala yangi terminlar

graph - grafik
given equation - berilgan tenglama
circle - aylana
xy-plane - koordinata tekisligi
point (a, b) - (a, b) nuqta
lies on the circle - aylanada yotadi
possible value - mumkin bo'lgan qiymat
center - markaz
radius - radius
coordinates - koordinatalar

29

Turbo-Module-2.23

Mark for Review

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A. The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

43, 45, 44, 43, 38, 39, 40, 46, 40

Enter your answer

29-masala yangi terminlar

data set - ma’lumotlar to‘plami
positive integers - musbat butun sonlar
less than 60 - 60 dan kichik
list of integers - butun sonlar ro‘yxati
mean - o‘rta arifmetik
integer mean - butun sonli o‘rtacha
greater than 42 - 42 dan katta
largest integer - eng katta butun son
data set A - A ma’lumotlar to‘plami
10 integers - 10 ta butun son

30 Turbo-Module-2.23

Mark for Review

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?

Enter your answer

30-masala yangi terminlar

quadratic function - kvadrat funksiya
models the depth - chuqurlikni modellashtiradi
depth - chuqurlik
below the surface - sirt ostida
seal - tyulen
minutes - daqiqalar
entered the water - suvga kirdi
dive - sho‘ng‘ish
maximum depth - eng katta chuqurlik
reached the surface - sirtga yetdi
estimated depth - taxminiy chuqurlik
nearest meter - eng yaqin metr